

Pigmented Lesions

Nevi

Benign growths

Hemangiomas

Skin Cancers

Case 1

- An otherwise healthy 8-year-old female presents for evaluation of a dome-shaped red-brown papule on the face that appeared 4 months ago and has grown rapidly. It is asymptomatic, and has not bled.



What is your differential diagnosis?

- Spitz nevus
- Halo nevus
- Blue nevus
- Nevus spilus

Spitz Nevus

- Round, well-circumscribed, dome-shaped, smooth-topped, hairless, small (<1 cm) papules or nodules
- Most often pink, tan, brown, sometimes black
- Often a history of **recent rapid growth** (arise within months)
- Most common location is the head and neck
- occur at all ages
 - 1/3 of patients are children
 - rarely seen > 40
- Histologic examination must be done to confirm the clinical diagnosis.
 - Spitz nevi are benign, but there can be a histologic similarity to melanoma.
 - Years ago these were called "juvenile melanoma."

Halo Nevus

- History: 3 stages
 - Development of a halo around a preexisting nevus, sometimes the halo may be preceded by faint erythema (redness)
 - Disappearance (months to years) of the nevus
 - Repigmentation (months to years) of the halo



Halo Nevus

- Course: the lesions undergo spontaneous involution, often with regression of the centrally located pigmented nevus
- Management: reassurance or excision if suspicious

Blue Nevus

- Acquired, asymptomatic, benign
- PE: 10mm, firm, round to oval, dark-blue to gray-black, sharply-defined papule
- commonly found on the dorsa of hands or feet, buttocks, back, scalp or face

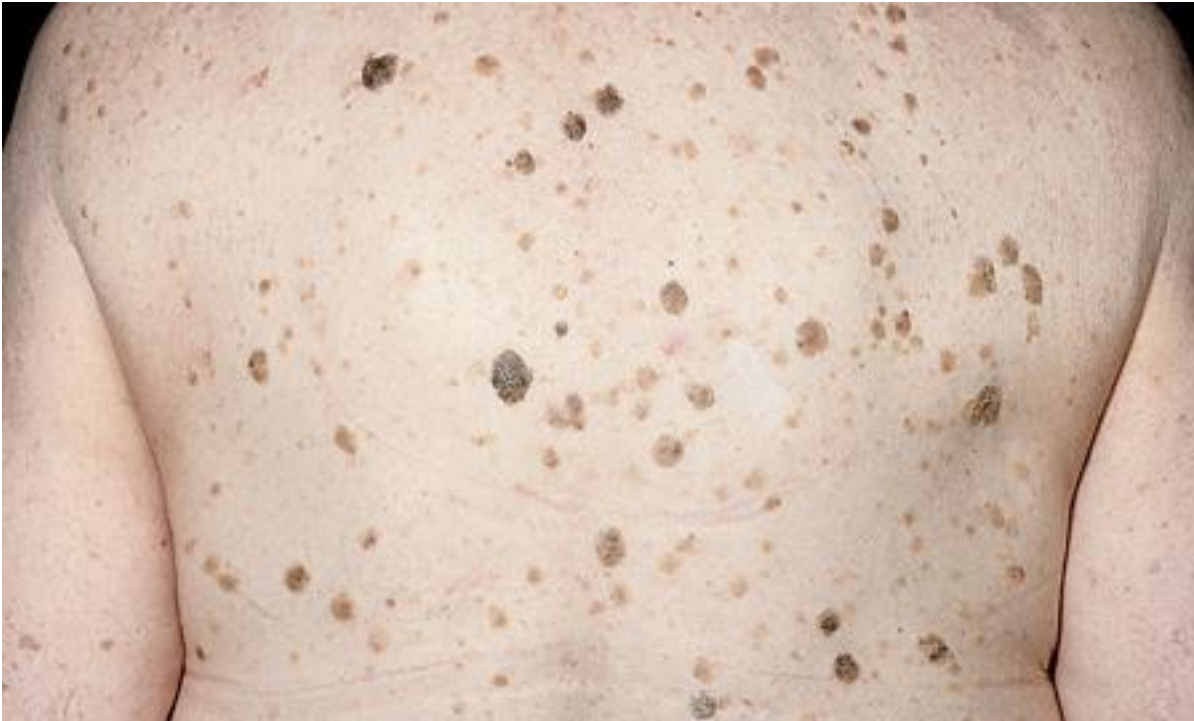


Nevus Spilus

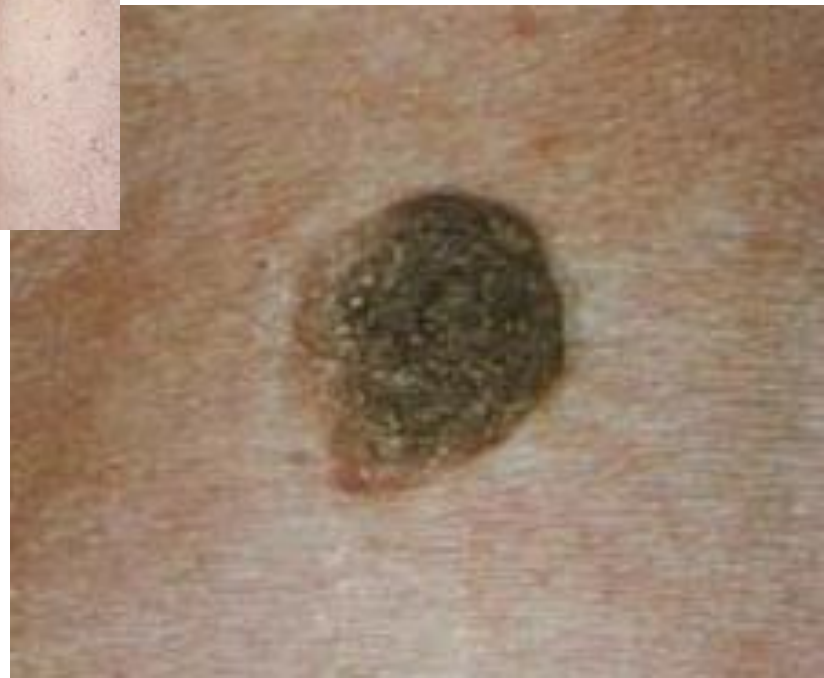
- light brown pigmented macule from a few cm to >15cm
- the distinctive feature of this lesion is the many dark brown small macules or papules (2mm) scattered throughout the pigmented background



Case 2



- discrete, “stuck-on”,
“greasy”, “warty”
brown, grey, tan



Seborrheic Keratosis

- The most common of the benign epithelial tumors.
- hereditary, begin to appear around age 30 and continue to occur over a lifetime, varying from a few to hundreds in some very elderly patients.
- Lesions are benign and do not require treatment except for cosmetic reasons.

Case 2a

- Pt states all of these lesions appeared abruptly over the last month. What is this called and why are you concerned?
- Sign of Leser-Trelat
- Internal malignancy (most commonly GI)

Seborrheic Keratosis

- ***dermatosis papulosa nigra***
- multiple tiny black lesions, some enlarging to more than a centimeter.
- seen in darkly pigmented pts.



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Case 3

1. 33 yo female with erythematous, slightly hyperpigmented dermal nodule on the leg
2. occurred after shaving



What's the sign?

- **“dimple sign”**-
lateral compression
with thumb and
index finger
produces a
depression or
dimple



C

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Dermatofibroma

- Very common
- Fibrous reaction to trauma or bite
- Legs>arms>trunk
- Important only because of its cosmetic appearance or its being mistaken for other lesions, such as malignant melanoma when it is pigmented.
- Usually asymptomatic papule or nodule, 3 to 10 mm in diameter.
- Firm, dome-shaped, or may be depressed
- Color is variable—skin-colored, pink, brown, tan, dark chocolate brown
- *Dimple sign*: lateral compression with thumb and index finger produces a depression or "dimple"

Case 4

- 64 yo obese female presents with these flesh colored pedunculated papules on her posterior and lateral neck that become irritated when she wears a necklace.
- She also has some under her armpits, breasts



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Acrochordons

- Syn: skin tags, cutaneous papilloma
- MC in middle aged & elderly, females, obese pts
- soft, skin-colored (tan, brown), oval, sessile, pedunculated papules
- Mc intertriginous, neck, eyelids, axilla, breasts
- Tx: scissor excision, electrodesiccation

Case 5

- 24 yo construction worker presents with this lesion that developed quickly after he injured his finger at work.
- Pt states it bleeds frequently



Pyogenic Granuloma

- Rapidly developing hemangioma arising at sites of minor trauma
- Bright red to violaceous, smooth, dome-shaped nodule
- bleeds spontaneously or after minor trauma
- Fingers, lips, mouth, trunk, toes
- Treatment is surgical excision with electrodesiccation at the base.
- The importance of **pyogenic granuloma** is that it can be mistaken for amelanotic nodular melanoma, and vice versa.

Pyogenic Granuloma



- What else is in your differential diagnosis for this lesion on the face of a child?

Spitz nevus

Case 6

- A 50 year old male presents with flesh to yellow colored papules on the cheeks
- They have a donut shaped appearance
- They are not painful but he is getting more and more of them over time
- they affect him cosmetically



Sebaceous hyperplasia

- Tumors of enlarged sebaceous glands
- Pale, yellow, donut, umbilicated papules
- Forehead, cheeks, lower lid, nose

What's your differential diagnosis?

syringomas

- Sweat duct tumor-benign adenoma
- 1-2mm flesh colored papules
- Multiple, **lower lids**, face, axilla, umbilicus, chest, vulva



What's your differential diagnosis?

trichoepithelioma

- Benign appendage tumors
- Occur at puberty
- Mc on face
- Pink to skin colored papules
- May appear similar to basal cell Ca



What's your differential diagnosis?

Milium

- 1-2 mm superficial white to yellow, keratin-containing epidermal cyst
- located on eyelids, cheek, forehead, & sites of trauma
- Occur at any age
- Tx: incision & expression of contents



Case 7

- A baby presents with a red strawberry appearing plaque on her trunk
- The parents are concerned because it is rapidly growing
- the dermatologist tells them not to worry as the lesion may disappear before she is 10 yo



Strawberry Nevus

- Benign **vascular tumor**/ proliferations
- Pale patch first month and **enlarges rapidly** by first year
- Red-purple, soft to firm
- Usually **involutates** by age 5 with a few by 10

What's your differential diagnosis?

Vascular malformations

Cavernous hemangioma

- deep, vascular malformation composed of capillary, lymphatic, & venous tissue
- soft, compressible blue-tinged, erythematous nodules

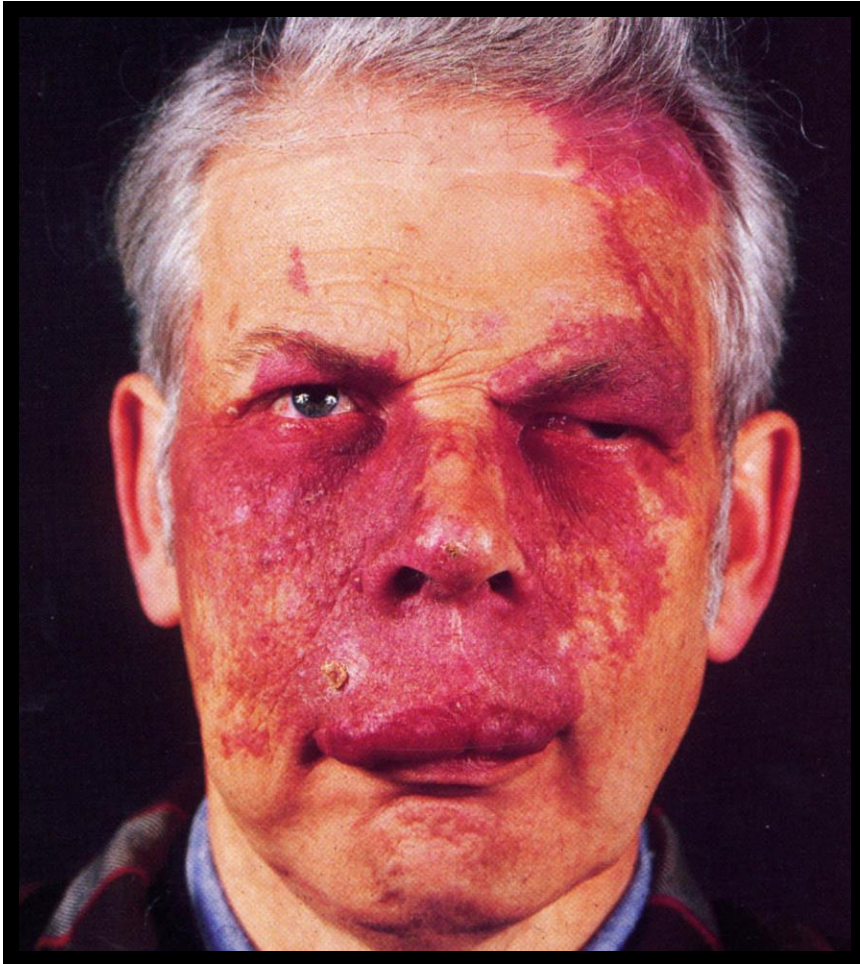


Port Wine Stain

- Vascular capillary malformations **that do not** undergo spontaneous involution
- irregularly shaped red to violaceous patch present at birth
- do not cross midline; dermatomal
- Usually confined to skin but may affect eye and leptomeninges



Port Wine Stain variants



- **Sturge-Weber Syndrome:**
port wine stain,
vascular malformations
of eye & leptomeninges

Port Wine Stain variants



- **Klippel-Trenaunay-Weber synd:** port wine with enlargement of extremity



Figure 1. Klippel-Trenaunay-Weber syndrome. (a) A large port wine stain on the upper arm and shoulder. (b) A small, dark, raised lesion on the arm. These findings are characteristic of Klippel-Trenaunay-Weber syndrome.

What's your differential diagnosis?

Cherry Angiomas

- common, asymptomatic, bright red domed papules
- principally on trunk
- first occur at 30 yrs & increase in number over years



Case 8

- 67 yr old male presents with these crusty lesions on his forehead and scalp. He describes continually picking at them and they keep coming back.



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Actinic Keratosis

- Common, sun-induced premalignant lesions
- Single, or multiple, discrete, rough, adherent, scaly papules
- Arise in areas of dermatoheliosis (head, neck, hands, sun damaged areas)
- Premalignant potential, may develop into SCC (squamous cell carcinoma), cutaneous horn
- Tx: cryotherapy, topical 5-fluorouracil (5-FU)

Case 9

- 61 yr old fisherman with this lesion, present for 1 year, slowly enlarging



B

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Case 9

- 46 yo runner with a slowly growing lesion on her nose for 1 yr



B

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Case 9

- 44 yr old female who reports frequent tanning habits presents with this lesion; unknown duration



A

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Basal Cell Carcinoma

- Most common type of skin cancer
- locally invasive & aggressive; limited capacity to metastasize
- Occur over age 40, pts w/poor tanning capacity & albinos
- 7 clinical types: **superficial, nodular, pigmented, ulcerative** (“rodent ulcer”), **sclerosing, cystic, basosquamous**
- Occurs on head, neck, face, canthi, nasolabial

Name that Basal Cell



Nodular BCC- pearly papule or nodule with telangiectasia, rolled border



A

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BCC, pigmented



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BCC, superficial



B

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BCC ulcerating

A large scar-like lesion, containing several small nodules of BCCs. On the lateral canthus of the eye there is a large ulcer with rolled borders representing rodent ulcer.



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Sclerosing (cicatricial) BCC-
waxy, white, indistinct, worst prognosis



Superficial BCC- red scaly plaque, pearly white border, resembles eczema, psoriasis, least aggressive



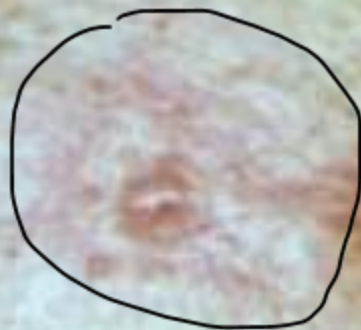
Basal Cell Carcinoma management

- excision
- electrodesiccation & curettage (ED&C)
- cryotherapy
- Mohs surgery
- topical immunomodulators
- radiation therapy

d&c



Before MOHS





Case 10

Describe this lesion
on the upper back of
a 54 yr old female

Lesion has been there
for 4 mos and didn't
respond to treatment
with topical
Hydrocortisone



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SCCIS (squamous cell carcinoma in situ)

Bowen's disease

- sharply demarcated, pink to red, scaling, papules, or plaques.
- SCCIS is most often caused by UVR (ultraviolet radiation) or HPV (human papilloma virus) infection.
- Commonly arises in lesions such as solar keratoses or HPV-induced squamous epithelial lesions
- Similar but usually nonscaly lesions on the glans and vulva are called *erythroplasia*.
- Anogenital HPV-induced SCCIS is referred to as *bowenoid papulosis*.
- Untreated SSCIS may progress to invasive SCC.
- Treatment: 5-fluorouracil, imiquimod, cryosurgery, CO₂ laser, or excision, including Mohs micrographic surgery.

Case 11

Describe these lesions



Invasive Squamous Cell Carcinoma

- Malignant tumor of keratinocytes
- SCC usually arises in epidermal precancerous lesions and, depending on etiology and level of differentiation, varies in its aggressiveness.
- Age of onset: > 55 yrs in U.S.
- indurated erythematous, hyperkeratotic, scaly papules, plaques, nodules, +/- erosions
- Occur on face, tips of ears, preauricular area, scalp, dorsal hands, forearms, lower lips
- Tx: cryotherapy, ED&C (debride and curette), Mohs, surgery
- The majority of UVR-induced lesions have a low rate of distant metastasis in otherwise healthy individuals. More aggressive SCC s occur in immunosuppressed pts

Case 12

- 55 yo healthy male
- 1 cm lesion on his back
- His wife wants you to biopsy it
- It has been present since birth
- He doesn't believe it has changed in appearance



Do you recommend biopsy?

A. Yes

B. No



Do you recommend biopsy?

A. Yes

B. No



What do you call this type of lesion?

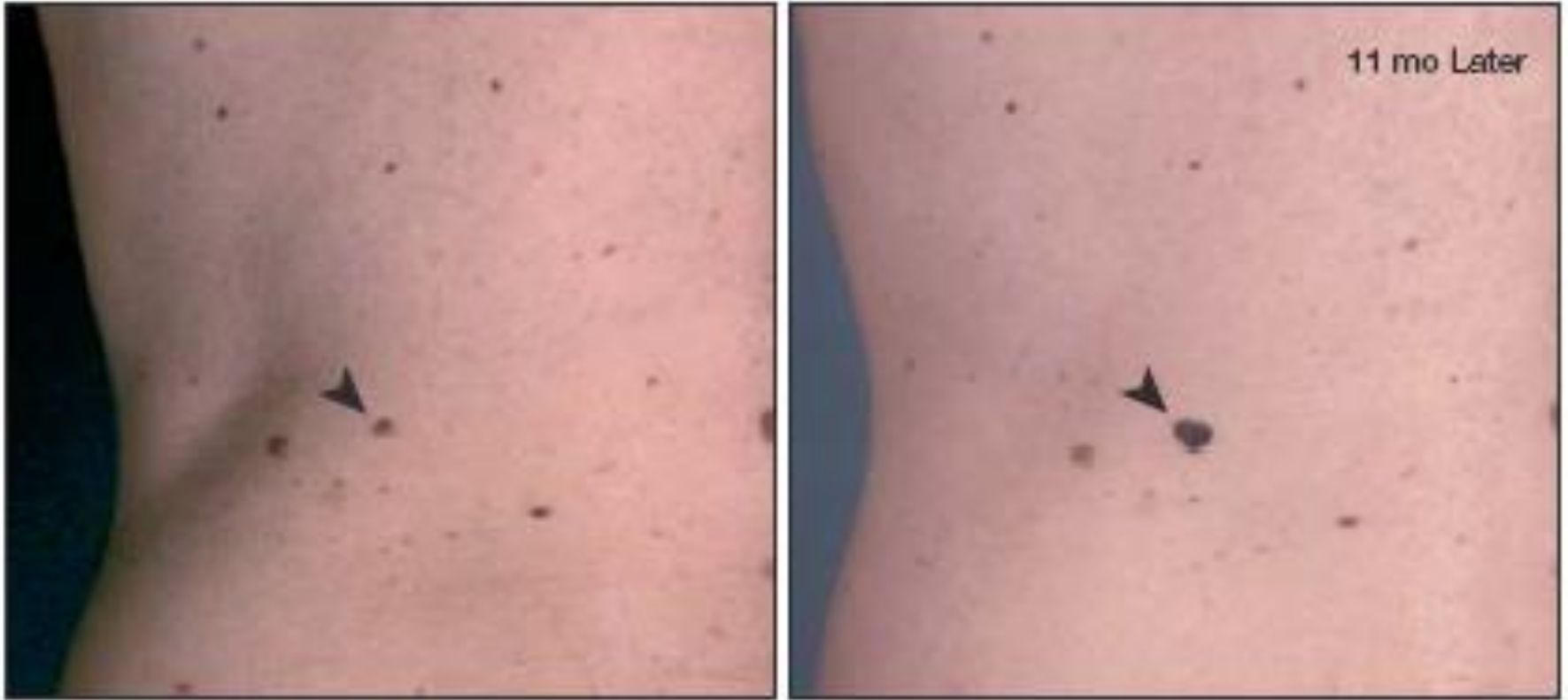
Congenital Nevus

- Mole present since birth
 - Highly pigmented
 - Hair growth in lesion
- Small < 1.5 cm, Medium 1.5-20 cm
 - Lifetime risk of melanoma equal to general population
- Giant nevi > 20 cm
 - Estimated lifetime risk of melanoma 3-8%
 - Prophylactic removal may be indicated
 - 40% of melanomas in kids develop within these nevi



Case 13

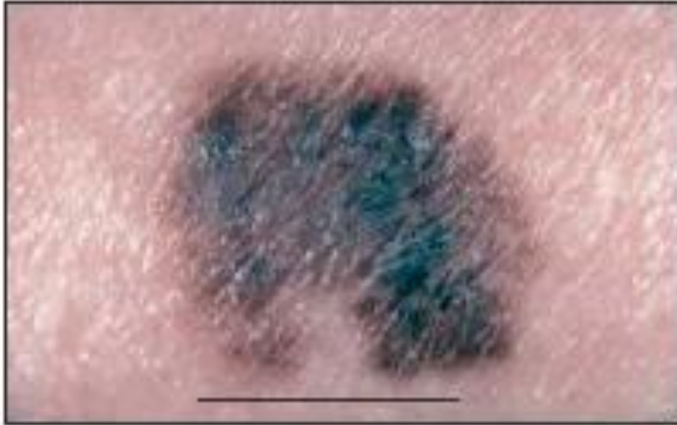
28 yo female presents with this lesion:



What next?

ABCDE's of Melanoma

Asymmetry



Border Irregularity



Color Variegation



Diameter > 6 mm



Evolving



What are the risk factors for Melanoma ?

- M-moles, atypical (dysplastic) >5
- M-moles, common >50
- R-red hair, freckling
- I-inability to tan, skin types I and II
- S-sunburn, severe especially <14 yo
- K-kindred, family hx of melanoma

Case 14



- 63 year old female shows you a dark area under her toenail
- She thinks she stubbed her toe a few months ago while working out in the garden
- She figured the dark area was just some bleeding

What's the sign?

Hutchinson's sign

Longitudinal pigmented streak that extends from proximal or lateral nail fold



Acral Lentiginous Melanoma

- Incidence equal in Caucasian & dark-skinned pts
- 70% of melanomas in African-Americans, Latinos, Asians
- Location of lesions
 - palms, soles, terminal phalanges, nail beds
 - mucous membranes
- poor prognosis
 - More advanced at time of diagnosis
 - Tend to metastasize early



- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>